Gaussian Processes

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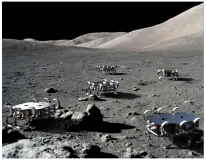
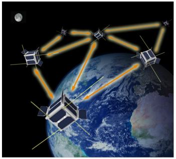
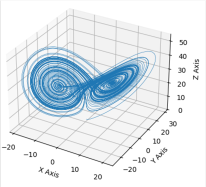
Motivation: Swarms and challenges

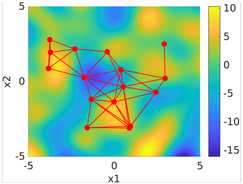

Rovers


Drones

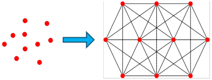

Satellites



Dynamics



Field Estimation


Control

Target tracking

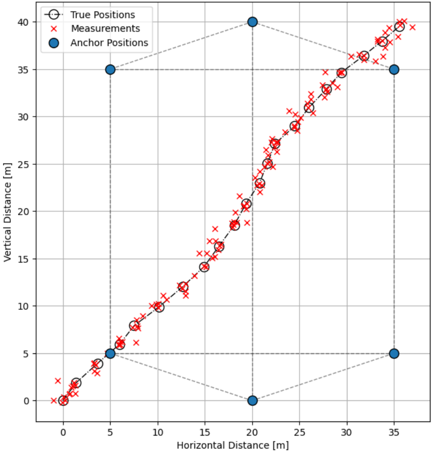


Figure: Single target tracking, using a set of anchors and distance measurements

Learning dynamics (1)

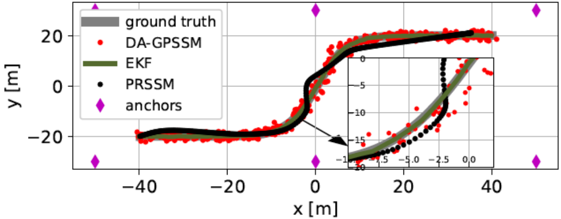


Figure: Learning non-linear dynamics using GPSSM

Mishra et.al., (2024) "Domain- Aware Reduced Rank Gaussian Process State Space Models"

Learning dynamics (2)

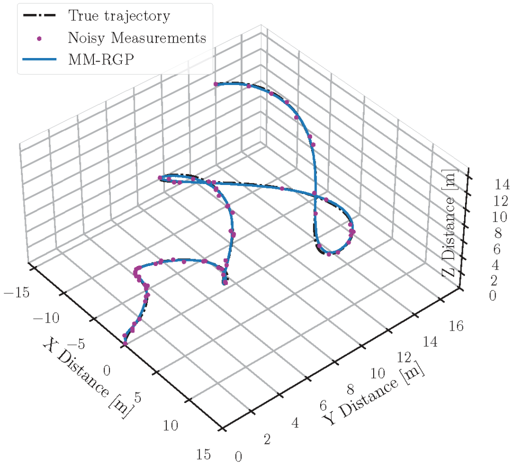


Figure: Learning non-linear dynamics using GP Mixture models

Balci et.al., (2025) "Multiple Model Recursive Gaussian Process for Robust Target Tracking"

Recap: Product rule and Bayes theorem

Given two random vectors $\mathbf{y}$ and $\mathbf{w}$ where

- $p(\mathbf{y}, \mathbf{w})$ is the joint PDF
- $p(\mathbf{y}|\mathbf{w})$ is the posterior PDF
- $p(\mathbf{y}), p(\mathbf{w})$ are the marginal PDFs of $\mathbf{y}$ and $\mathbf{w}$

- Product rule

$$p(\mathbf{y}, \mathbf{w}) = p(\mathbf{y}|\mathbf{w})p(\mathbf{w}) = p(\mathbf{w}|\mathbf{y})p(\mathbf{y})$$

- Bayes theorem

$$p(\mathbf{y}|\mathbf{w}) = \frac{p(\mathbf{y}, \mathbf{w})}{p(\mathbf{w})} = \frac{p(\mathbf{w}|\mathbf{y})p(\mathbf{y})}{p(\mathbf{w})}$$

Bayesian Estimation

Consider a data vector $\mathbf{y}$ and unknown weights $\mathbf{w}$, whose distributions are

$$\mathbf{y} \sim \ p(\mathbf{y}), \quad \mathbf{w} \sim \ p(\mathbf{w}) \tag{1}$$

then, the optimal estimator $\hat{\mathbf{w}} \equiv \hat{\mathbf{w}}(\mathbf{y})$ given $\mathbf{y}$, that minimizes the BMSE i.e.,

$$\mathrm{BMSE}(\hat{\mathbf{w}}) = \mathbb{E}_{\mathbf{y},\mathbf{w}}[\|\hat{\mathbf{w}} - \mathbf{w}\|^2] = \int_y \int_w \|\hat{\mathbf{w}} - \mathbf{w}\|^2 p(\mathbf{y}, \mathbf{w}) d\mathbf{w} d\mathbf{y}$$

is the MMSE estimator (obtained via the Bayes theorem) i.e.,

$$\hat{\mathbf{w}} = \int \mathbf{w} p(\mathbf{w}|\mathbf{y}) d\mathbf{w} = \mathbb{E}(\mathbf{w}|\mathbf{y}) = \boldsymbol{\mu}_{w|y},$$

which is also the MAP and LMMSE estimator, if $p(.)$ in (1) is Gaussian.

Multivariate Gaussian (1)

Consider $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$ and $\mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w)$, then we have

Prior

$$p(\mathbf{w}) = \frac{1}{\sqrt{(2\pi)^L \det(\boldsymbol{\Sigma}_w)}} \exp\left[-\frac{1}{2}\left[\mathbf{w} - \boldsymbol{\mu}_w\right]^T \boldsymbol{\Sigma}_w^{-1}\left[\mathbf{w} - \boldsymbol{\mu}_w\right]\right]$$

Marginal

$$p(\mathbf{y}) = \frac{1}{\sqrt{(2\pi)^M \det(\boldsymbol{\Sigma}_y)}} \exp\left[-\frac{1}{2}\left[\mathbf{y} - \boldsymbol{\mu}_y\right]^T \boldsymbol{\Sigma}_y^{-1}\left[\mathbf{y} - \boldsymbol{\mu}_y\right]\right]$$

Joint Gaussian distribution

$$p(\mathbf{y}, \mathbf{w}) = \frac{1}{\sqrt{(2\pi)^{M+L} \det(\bar{\boldsymbol{\Sigma}})}} \exp\left[-\frac{1}{2}\begin{bmatrix}\mathbf{y} - \boldsymbol{\mu}_y \\ \mathbf{w} - \boldsymbol{\mu}_w\end{bmatrix}^T \bar{\boldsymbol{\Sigma}}^{-1}\begin{bmatrix}\mathbf{y} - \boldsymbol{\mu}_y \\ \mathbf{w} - \boldsymbol{\mu}_w\end{bmatrix}\right]$$

where the moments are

$$\mathbb{E}\left(\begin{bmatrix}\mathbf{y} \\ \mathbf{w}\end{bmatrix}\right) = \begin{bmatrix}\boldsymbol{\mu}_y \\ \boldsymbol{\mu}_w\end{bmatrix}, \quad \bar{\boldsymbol{\Sigma}} = \begin{bmatrix}\boldsymbol{\Sigma}_y & \boldsymbol{\Sigma}_{yw} \\ \boldsymbol{\Sigma}_{yw}^T & \boldsymbol{\Sigma}_w\end{bmatrix}$$

Multivariate Gaussian (2)

Conditional density is

$$p(\mathbf{w}|\mathbf{y}) = \frac{p(\mathbf{y}, \mathbf{w})}{p(\mathbf{y})} = \mathcal{N}(\boldsymbol{\mu}_{w|y}, \boldsymbol{\Sigma}_{w|y})$$

where

$$\begin{aligned} \boldsymbol{\mu}_{w|y} &= \boldsymbol{\mu}_w + \boldsymbol{\Sigma}_{yw}^T \boldsymbol{\Sigma}_y^{-1}(\mathbf{y} - \boldsymbol{\mu}_y) \\ \boldsymbol{\Sigma}_{w|y} &= \boldsymbol{\Sigma}_w - \boldsymbol{\Sigma}_{yw}^T \boldsymbol{\Sigma}_y^{-1} \boldsymbol{\Sigma}_{yw} \end{aligned}$$

Now, consider the simple measurement model

$$\mathbf{y} = \mathbf{w} + \boldsymbol{\epsilon}, \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}),$$

then the moments of the conditional PDF are

$$\begin{aligned} \boldsymbol{\mu}_{w|y} &= \boldsymbol{\mu}_w + \boldsymbol{\Sigma}_w(\boldsymbol{\Sigma}_w + \sigma_\epsilon^2 \mathbf{I})^{-1}(\mathbf{y} - \boldsymbol{\mu}_w) \\ \boldsymbol{\Sigma}_{w|y} &= \boldsymbol{\Sigma}_w - \boldsymbol{\Sigma}_w(\boldsymbol{\Sigma}_y + \sigma_\epsilon^2 \mathbf{I})^{-1} \boldsymbol{\Sigma}_y \end{aligned}$$

Bayesian Gauss Markov Model

Consider the Bayesian Gauss Markov Model

$$\mathbf{y} = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\epsilon}, \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_w, \mathbf{\Sigma}_w), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}),$$

where $\mathbf{y}$ is $M \times 1$ and $\mathbf{w}$ is $L \times 1$ then the joint Gaussian moments are

$$\mathbb{E}\left(\begin{bmatrix}\mathbf{y}\\ \mathbf{w}\end{bmatrix}\right) = \begin{bmatrix}\mathbf{\Phi}\boldsymbol{\mu}_w\\ \boldsymbol{\mu}_w\end{bmatrix}, \quad \bar{\mathbf{\Sigma}} = \begin{bmatrix}\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I} & \mathbf{\Phi}\mathbf{\Sigma}_w\\ \mathbf{\Sigma}_w\mathbf{\Phi}^T & \mathbf{\Sigma}_w\end{bmatrix}$$

such that

$$p(\mathbf{y},\mathbf{w}) = \frac{1}{\sqrt{(2\pi)^{M+L}\det(\bar{\mathbf{\Sigma}})}}\exp\left[-\frac{1}{2}\begin{bmatrix}\mathbf{y}-\mathbf{\Phi}\boldsymbol{\mu}_w\\ \mathbf{w}-\boldsymbol{\mu}_w\end{bmatrix}^T \bar{\mathbf{\Sigma}}^{-1}\begin{bmatrix}\mathbf{y}-\mathbf{\Phi}\boldsymbol{\mu}_w\\ \mathbf{w}-\boldsymbol{\mu}_w\end{bmatrix}\right]$$

then the conditional PDF is also Gaussian i.e., $p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_{w|y}, \mathbf{\Sigma}_{w|y})$ where

$$\begin{aligned}
\boldsymbol{\mu}_{w|y} &= \boldsymbol{\mu}_w + \mathbf{\Sigma}_w\mathbf{\Phi}^T(\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I})^{-1}(\mathbf{y} - \mathbf{\Phi}\boldsymbol{\mu}_w)\\
\mathbf{\Sigma}_{w|y} &= \mathbf{\Sigma}_w - \mathbf{\Sigma}_w\mathbf{\Phi}^T(\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{\Phi}\mathbf{\Sigma}_w
\end{aligned}$$

Bayesian Gauss Markov Model: Special case

Consider (a special case of) the Bayesian Gauss Markov Model, when $\mathbb{E}[\mathbf{w}] = \mathbf{0}$

$$\mathbf{y} = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\epsilon}, \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_w), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}),$$

where $\mathbf{y}$ is $M \times 1$ and $\mathbf{w}$ is $L \times 1$ then the joint covariance is

$$\bar{\mathbf{\Sigma}} = \begin{bmatrix} \mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I} & \mathbf{\Phi}\mathbf{\Sigma}_w \\ \mathbf{\Sigma}_w\mathbf{\Phi}^T & \mathbf{\Sigma}_w \end{bmatrix}$$

such that

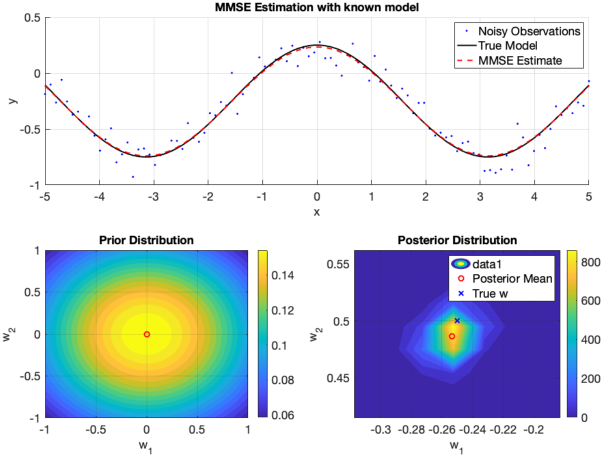
$$p(\mathbf{y}, \mathbf{w}) = \frac{1}{\sqrt{(2\pi)^{M+L}\det(\bar{\mathbf{\Sigma}})}} \exp\left[-\frac{1}{2}\begin{bmatrix}\mathbf{y}\\ \mathbf{w}\end{bmatrix}^T \bar{\mathbf{\Sigma}}^{-1}\begin{bmatrix}\mathbf{y}\\ \mathbf{w}\end{bmatrix}\right]$$

then the moments of the conditional PDF $p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_{w|y}, \mathbf{\Sigma}_{w|y})$ simplify to

$$\begin{aligned} \boldsymbol{\mu}_{w|y} &= \mathbf{\Sigma}_w\mathbf{\Phi}^T(\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y} \\ \mathbf{\Sigma}_{w|y} &= \mathbf{\Sigma}_w - \mathbf{\Sigma}_w\mathbf{\Phi}^T(\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{\Phi}\mathbf{\Sigma}_w \end{aligned}$$

Example

An example with $\mathbf{\Phi} = \begin{bmatrix} \mathbf{1} & \cos(\boldsymbol{x}) \end{bmatrix}$, $\mathbf{w} = [-0.25, 0.5]^T$, $\sigma_\epsilon^2 = 0.01$



Bayesian Gauss Markov Model: Prediction

Consider (a special case of) the Bayesian Gauss Markov Model, when $\mathbb{E}[\mathbf{w}] = \mathbf{0}$

$$\mathbf{y} = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\epsilon}, \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_w), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}),$$

and given a new input

$$y_* = \boldsymbol{\phi}_*^T \mathbf{w} + \epsilon_*$$

the joint distribution is

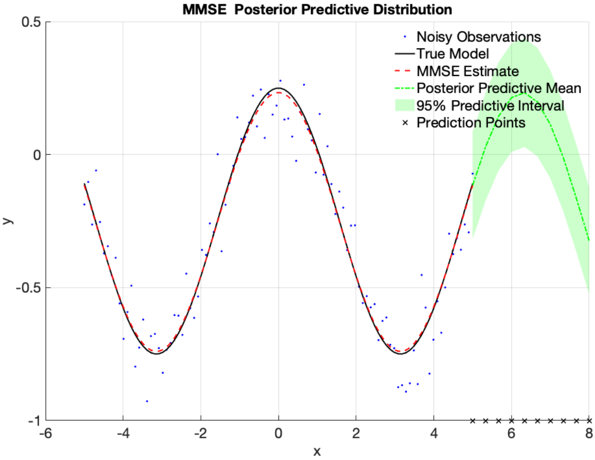
$$\begin{bmatrix} \mathbf{y} \\ y_* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2 \mathbf{I} & \mathbf{\Phi}\mathbf{\Sigma}_w \boldsymbol{\phi}_* \\ \boldsymbol{\phi}_*^T \mathbf{\Sigma}_w \mathbf{\Phi}^T & \boldsymbol{\phi}_*^T \mathbf{\Sigma}_w \boldsymbol{\phi}_* + \sigma_\epsilon^2 \end{bmatrix} \right)$$

then the predictive mean is

$$\mathbb{E}[y_*|\mathbf{y}] = \boldsymbol{\phi}_*^T \mathbf{\Sigma}_w \mathbf{\Phi}^T (\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}$$

Example (2)

Predictions at $\phi_* = \begin{bmatrix} 1 & \cos(x_*) \end{bmatrix}$, where $x_* \in (5, 8)$



Basis functions

- Consider a dataset $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\} \equiv \{\mathbf{x}_m, y_m\}_{m=1}^{M}$ where $\mathbf{x}_m \in \mathcal{X}$ denotes the input vectors in D-dimensional space e.g., $\mathbb{R}^D$, and y_m is the corresponding scalar output.
- Can we generalize Bayesian Gauss Markov to feature spaces i.e.,

$$y_m = f(\mathbf{x}_m) + \epsilon_m \quad \text{where } f(\mathbf{x}_m) = \boldsymbol{\phi}(\mathbf{x}_m)^T \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_w)$$

  where $\phi(\cdot)$ is a basis function
- Observe that for any pair of input vectors $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$

$$\begin{aligned} \mathbb{E}[f(\mathbf{x})] &= \boldsymbol{\phi}(\mathbf{x})^T \mathbb{E}[\mathbf{w}] = \mathbf{0} \\ \mathbb{E}[f(\mathbf{x})f(\mathbf{x}')] &= \boldsymbol{\phi}(\mathbf{x})^T \mathbb{E}[\mathbf{w}\mathbf{w}^T] \boldsymbol{\phi}(\mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\Sigma}_w \boldsymbol{\phi}(\mathbf{x}') \end{aligned}$$

- Hence, $f(\mathbf{x})$ and $f(\mathbf{x}')$ are jointly Gaussian with zero mean and covariance given by $\kappa(\mathbf{x}, \mathbf{x}') \equiv \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\Sigma}_w \boldsymbol{\phi}(\mathbf{x}')$.

Definitions

Definition (Gaussian Process)

Let $\boldsymbol{f} = \big[f(\mathbf{x}_1), f(\mathbf{x}_2), \ldots, f(\mathbf{x}_M)\big]$ denote the function values at a set of known inputs $\mathbf{X} = \{\mathbf{x}_m \in \mathcal{X}\}_{m=1}^{M}$, where $f(\cdot) : \mathcal{X} \to \mathbb{R}$ be a real valued function evaluated at these points. If $\boldsymbol{f}$ is jointly Gaussian $\forall M$, then $f(\cdot)$ is a Gaussian Process or GP.

Definition (GP Mean and Covariance)

A GP can be completely defined by its mean $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$, and a positive definite covariance function $\mathcal{K}(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))^T]$ i.e.,

$$f(\mathbf{x}) \sim \mathcal{GP}\big(m(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}')\big) \quad \{\mathbf{x}, \mathbf{x}' \in \mathcal{X}\}$$

Given $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_M]$, we have $p(\boldsymbol{f}|\mathbf{X}) = \mathcal{N}(\boldsymbol{f}|\boldsymbol{\mu}_X, \mathbf{K}_{XX})$ where

$$\boldsymbol{\mu}_X \equiv [m(\mathbf{x}_1), \ldots, m(\mathbf{x}_M)]^T \quad (= \mathbf{0} \text{ assumed in this lecture})$$

$$[\mathbf{K}_{XX}]_{i,j} \equiv \kappa(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\Sigma}_w \boldsymbol{\phi}(\mathbf{x}')$$

Bayesian inference: Gaussian Process

Let $f(\mathbf{x}_m) = \boldsymbol{\phi}(\mathbf{x}_m)^T\mathbf{w}$, then consider the data model

$$y_m = f(\mathbf{x}_m) + \epsilon_m \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_w), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I}),$$

where the prior on the weights is $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_w)$. Populating M data points

$$\mathbf{y} = \boldsymbol{\Phi}\mathbf{w} + \boldsymbol{\epsilon}, \quad \text{where} \quad \boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\phi}(\mathbf{x}_1)^T \\ \boldsymbol{\phi}(\mathbf{x}_2)^T \\ \dots \\ \boldsymbol{\phi}(\mathbf{x}_M)^T \end{bmatrix} \quad \text{and} \quad \text{Cov}(\mathbf{y}) = \boldsymbol{\Phi}\boldsymbol{\Sigma}\boldsymbol{\Phi}^T + \sigma_\epsilon^2\mathbf{I}$$

Thus for a new data point $\mathbf{x}_*$,

$$\begin{aligned} f(\mathbf{x}_*) \equiv f_* &= \boldsymbol{\phi}(\mathbf{x}_*)^T\mathbf{w}, \\ \text{Var}(f_*) &= \boldsymbol{\phi}(\mathbf{x}_*)^T\boldsymbol{\Sigma}_w\boldsymbol{\phi}(\mathbf{x}_*), \\ \text{Cov}(\mathbf{y}, f_*) &= \boldsymbol{\Phi}\boldsymbol{\Sigma}_w\boldsymbol{\phi}(\mathbf{x}_*) \end{aligned}$$

Can we write the joint distribution $p(\mathbf{y}, f_*)$ in terms of the known moments ?

Bayesian inference: Gaussian Process (2)

Joint Distribution

$$\begin{bmatrix}\mathbf{y}\\ f_*\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\mathbf{0}\\ 0\end{bmatrix}, \begin{bmatrix}\mathbf{\Phi}\mathbf{\Sigma}_w\mathbf{\Phi}^T + \sigma_\epsilon^2\mathbf{I} & \mathbf{\Phi}\mathbf{\Sigma}_w\boldsymbol{\phi}(\mathbf{x}_*)\\ \boldsymbol{\phi}^T(\mathbf{x}_*)\mathbf{\Sigma}_w\mathbf{\Phi}^T & \boldsymbol{\phi}^T(\mathbf{x}_*)\mathbf{\Sigma}_w\boldsymbol{\phi}(\mathbf{x}_*)\end{bmatrix}\right)$$

Consider the definitions

$$\mathbf{K}_{XX} = \begin{bmatrix}\kappa(\mathbf{x}_1,\mathbf{x}_2) & \dots & \kappa(\mathbf{x}_1,\mathbf{x}_M)\\ \vdots & \dots & \vdots\\ \kappa(\mathbf{x}_M,\mathbf{x}_1) & \dots & \kappa(\mathbf{x}_M,\mathbf{x}_M)\end{bmatrix}, \quad \boldsymbol{\kappa}_* = \begin{bmatrix}\kappa(\mathbf{x}_1,\mathbf{x}_*)\\ \kappa(\mathbf{x}_2,\mathbf{x}_*)\\ \dots\\ \kappa(\mathbf{x}_M,\mathbf{x}_*)\end{bmatrix}, \quad \kappa_{**} \equiv \kappa(\mathbf{x}_*, \mathbf{x}$$

then the joint distribution is

$$\begin{bmatrix}\mathbf{y}\\ f_*\end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix}\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I} & \boldsymbol{\kappa}_*\\ \boldsymbol{\kappa}_*^T & \kappa_{**}\end{bmatrix}\right)$$

and the conditional mean and variance are

$$\hat{f}_{*|y} \equiv \mathbb{E}[f_*|\mathbf{y}] = \boldsymbol{\kappa}_*^T(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y}$$
$$\sigma_{*|y} \equiv \mathrm{Var}[f_*|\mathbf{y}] = \kappa_{**} - \boldsymbol{\kappa}_*^T(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}\boldsymbol{\kappa}_*$$

Kernels

The kernel is a function, which specifes the covariance between values of a random functions $f(\mathbf{x})$ and $(\mathbf{x}')$ in Gaussian Process

$$[\mathbf{K}_{XX}]_{i,j} \equiv \kappa(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\Sigma}_w \boldsymbol{\phi}(\mathbf{x}') : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$$

Kernel Properties

- **Symmetry**: $\kappa(\mathbf{x}, \mathbf{x}') = \kappa(\mathbf{x}', \mathbf{x})$
- **Positive semi-definite**: $\mathbf{z}^T \mathbf{K}_{XX} \mathbf{z} \geq 0$
- Encodes similarity between inputs

Common Kernels

- Linear: $\kappa(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}'$
- Squared Exponential kernel:

$$\kappa(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left[-\frac{1}{2\ell^2} \|\mathbf{x} - \mathbf{x}'\|_2^2 \right] \quad \mathbf{x}, \mathbf{x}' \in \mathcal{X},$$

which contain the hyperparameters $[\sigma_f^2,\ \ell]$ i.e., the signal power (σ_f^2) and length-scale (ℓ) respectively.

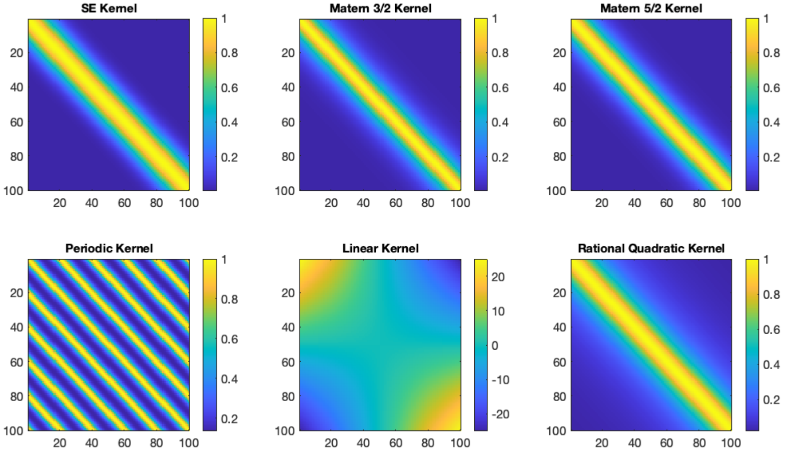


Figure: Visualization of Gaussian Process Kernel Matrices: Canonical forms

see https://smlbook.org/GP/

Bayesian inference: Gaussian Process

- Consider the Squared Exponential kernel i.e.,

$$\kappa(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left[-\frac{1}{2\ell^2}\|\mathbf{x} - \mathbf{x}'\|_2^2\right] \quad \mathbf{x}, \mathbf{x}' \in \mathcal{X},$$

  which contain the hyperparameters $[\sigma_f^2,\ \ell]$ i.e., the signal power (σ_f^2) and length-scale (ℓ) respectively.

- Given the dataset $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$, the predictive distribution at test input $\mathbf{x}_*$ is a joint Gaussian distribution conditioned on the given information i.e.,

$$p(f_*|\mathbf{x}_*, \mathcal{D}, \boldsymbol{\theta}) \sim \mathcal{N}(\hat{f}_{*|\mathcal{D}}, \boldsymbol{\Sigma}_{*|\mathcal{D}})),$$

  where $\boldsymbol{\theta} = [\sigma_f^2,\ \ell, \sigma_\epsilon]^T$

$$\begin{aligned} \hat{f}_{*|\mathcal{D}} &= \boldsymbol{\kappa}_{X*}^T(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y}, \\ \sigma_{*|\mathcal{D}} &= \kappa_{**} - \boldsymbol{\kappa}_{X*}^T(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}\boldsymbol{\kappa}_{X*}. \end{aligned}$$

  and $\mathbf{K}_{XX}, \boldsymbol{\kappa}_{X*}$ and κ_{**} are the kernel relationships between $\mathbf{X}$ and $\mathbf{X}$, $\mathbf{X}$ and $\mathbf{x}_*$, and $\mathbf{x}_*$ and $\mathbf{x}_*$, respectively.

Hyperparameter optimization

Given the dataset $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$, recollect our data model

$$\mathbf{y} = f(\mathbf{x}) + \boldsymbol{\epsilon} \quad \text{where } f(\mathbf{x}) \sim \mathcal{GP}(\mathbf{0}, \kappa(\mathbf{x}, \mathbf{x}')), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I})$$

Note that the Marginal Likelihood is

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K} + \sigma_\epsilon^2 \mathbf{I})$$

which for M data points is

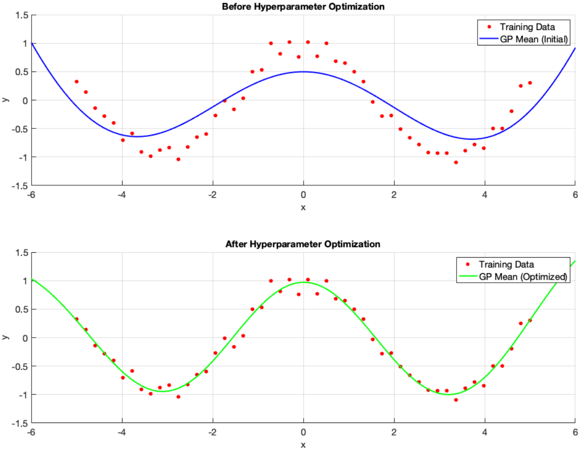
$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^M \det(\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})}} \exp\left[-\frac{1}{2}\mathbf{y}^T (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}\right]$$

and the corresponding Marginal Log Likelihood (MLL) is

$$\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = -\frac{1}{2}\left[\mathbf{y}^T (\mathbf{K}_{XX} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y} + \log\det(\mathbf{K}_{XX} + \sigma_\epsilon^2 \mathbf{I}) + N \log(2\pi)\right]$$

Solve for $\boldsymbol{\theta} = [\sigma_f^2,\ \ell, \sigma_\epsilon^2] \rightarrow$ Maximize MLL or Minimize Negative MLL

Example: Hyperparameter optimization



(Before) $\boldsymbol{\theta} = [\sigma_s^2 = 2, \ell = 5, \sigma_\epsilon^2 = 0.2]$, (After) $\boldsymbol{\theta} = [\sigma_s^2 = 1, \ell = 2, \sigma_\epsilon^2 = 0.1]$

Hyperparameter optimization (2)

Recollect the Negative Marginal Log-Likelihood of the data $p(\mathbf{y}|\mathbf{X},\boldsymbol{\theta})$ is

$$\begin{aligned}\log p(\mathbf{y}|\mathbf{X},\boldsymbol{\theta}) &= -\frac{1}{2}\left[\mathbf{y}^T(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y} + \log\det(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I}) + N\log(2\pi)\right] \\ &\propto -\frac{1}{2}\left[\mathbf{y}^T(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y} + \log\det(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I})\right]\end{aligned}$$

Observe the 2 key components

1. $\mathbf{y}^T(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y} \rightarrow$ Squared Mahalanobis distance
2. $\log\det(\mathbf{K}_{XX}+\sigma_\epsilon^2\mathbf{I}) \rightarrow$ Model complexity

Computational complexity

Recollect, that given $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$ solving the Negative Marginal Log Likelihood

$$-\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \propto -\frac{1}{2}\left[\mathbf{y}^T(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}\mathbf{y} + \log\det(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})\right]$$

has a computational bottleneck in calculating

- $\log\det(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})$
- $(\mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I})^{-1}$

which typically requires $\mathcal{O}(M^3)$

How to overcome the growing dimensionality of $\tilde{\mathbf{K}} \equiv \mathbf{K}_{XX} + \sigma_\epsilon^2\mathbf{I}$?
Approaches to overcome this limitation:

a Recursive/Iterative updates

b Divide dataset $\mathcal{D}$ (Distributed GPR, Last lecture)

c Low-rank approximations of Kernel structure

. $\vdots$

Nystrom Approximation

Nystrom Method:

Choose a small subset of inducing points $\mathbf{U} = \{\mathbf{u}_m\}_{m=1}^{M'}$, with $M' \ll M$

Approximate the full kernel $\tilde{\mathbf{K}}_{XX}$

$$\tilde{\mathbf{K}}_{XX} \approx \mathbf{K}_{XU}\mathbf{K}_{UU}^{-1}\mathbf{K}_{UX}$$

where

$\mathbf{K}_{XU}$: kernel between training points and inducing points

$\mathbf{K}_{UU}$: kernel between inducing points

Key Features:

Works directly with a subset of inducing points

Low-rank approximation: rank$(\tilde{\mathbf{K}}^N) \leq M'$

Accuracy depends on choice of inducing points

Scalable Kernel Interpolation (SKI)

SKI Method:

For each training input $\mathbf{x}_i$, compute interpolation weights $\{w_{i,m}\}_{m=1}^{M'}$ for grid points $\mathbf{U}$:

$$\kappa(\mathbf{x}_i, \mathbf{u}_j) \approx \sum_{m=1}^{M'} w_{i,m} \kappa(\mathbf{u}_m, \mathbf{u}_j)$$

Form the interpolation matrix $\mathbf{W} \in \mathbb{R}^{M \times M'}$ such that $\mathbf{K}_{XU} \approx \mathbf{W}\mathbf{K}_{UU}$.

Kernel approximation:

$$\mathbf{K}_{XX} \approx \tilde{\mathbf{K}}^{SKI} = \mathbf{W}\mathbf{K}_{UU}\mathbf{W}^T$$

Key Features:

Uses interpolation weights for flexible approximation

Exploits grid structure of inducing points for fast computations

$\mathbf{W}$ is often sparse (memory-efficient), not strictly low-rank

Observe: In Nystrom, $\mathbf{K}_{XU}$ is directly computed from the kernel, $\mathbf{W}$ is a way to interpolate $\mathbf{K}_{XU}$ from $\mathbf{K}_{UU}$

Example: Structured Kernal Interpolation

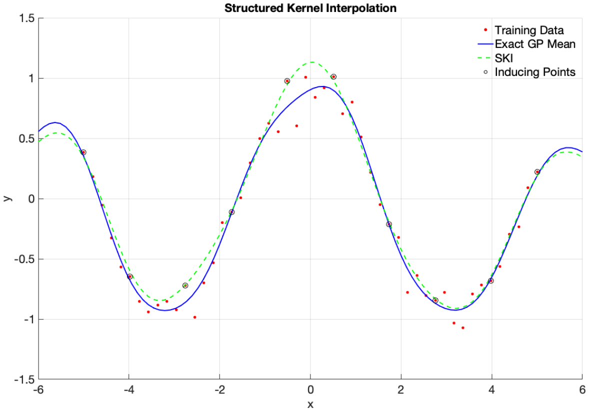


Figure: $M = 50, M' = 10, \sigma_\epsilon^2 = 0.1$

Summary

References:

- (Chapter 1-2), Williams, Christopher KI, and Carl Edward Rasmussen. Gaussian processes for machine learning. Cambridge, MA: MIT press, 2006.
- (Chapter 6), Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4. New York: springer, 2006.

Update schedule:

- Wednesday, March 11: **No lecture**
- Wednesday, March 18: Intro to Graph + Distributed Kalman Filtering
- Wednesday, March 25: Distributed Gaussian Processes